Group II Row IV: DGLAP, running coupling and finite terms, step by step

Your equation (Group II Row IV, no c.c.). $a \equiv \frac{k^+}{\vee}$, $\alpha_s = \frac{g^2}{4\pi}$, $d_\perp = 2-\epsilon$, $\beta_0 = \frac{11}{3}N_c$ ($n_f = 0$), $c_0 = 2e^{-\gamma_E}$, $\Delta \equiv y' - y$, $\ell_\Delta(\mu) \equiv \log\frac{\Delta^2\mu^2}{c_0^2} = 2\gamma_E + \log\frac{(y'-y)^2\mu^2}{4}$.

1 Answer

- **DGLAP.** The pole of the row is $\frac{\alpha_s}{2\pi}\big(-\frac{2}{\epsilon}\big)P^{(r)}_{gg,+}\otimes\mathrm{LO}$, where $P^{(r)}_{gg,+}$ is the real part of P_{gg}. Write $P^{(r)}_{gg,+} = P_{gg} - \frac{\beta_0}{2}\delta(1-\xi)$.
  - The P_{gg} part is the DGLAP pole. It is absorbed in the fragmentation function $D_{h/g}(\zeta,\mu_F)$, which then obeys $\mu_F^2\partial_{\mu_F^2}D = \frac{\alpha_s}{2\pi}P_{gg}\otimes D$.
  - The $-\frac{\beta_0}{2}\delta(1-\xi)$ part is left in the row.
- **Running coupling.** The $\frac{\beta_0}{2}$ term left in the row is $+\frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}[\frac{2}{\epsilon}+\ell_\Delta(\mu)]$. Its pole is cancelled exactly by the $\overline{MS}$ counterterm of the coupling of LO. What remains turns the coupling of LO into a running one:

$$\mathrm{LO}\big|_{\alpha_s(\mu_R)} + \frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}\big[\ell_\Delta(\mu_R)\big] = \mathrm{LO}\Big[\alpha_s\Big(\frac{c_0}{|y'-y|}\Big)\Big] + O(\alpha_s^3).$$

  The coupling runs at the transverse distance $|y'-y|$, which is conjugate to $\mathbf{k}$ through $e^{-ik\cdot(y'-y)}$. μ_R drops out.
- **Finite terms of the row:**
  - the DGLAP coefficient $-\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}P_{gg}(\xi)\,\mathrm{LO}_\xi[\ell_\Delta(\mu_F)]$;
  - $-\log\frac{\vee}{k^+}H_4$, from (L2)+(L3);
  - the linearly polarized line (L5) at $d_\perp = 2$.

  Eq. (4.2) gives all of them in your notation.

Other β_0 terms. Group I Row II part 1 ($\frac{11}{3}$) and Group I Row V $d^3N_{2b\,\mathrm{UV}}$ ($-\frac{11}{6}$) have β_0 terms of their own. These cancel each other exactly, including their poles and μ (`Other_terms_analysis.pdf`, Sec. 5). Keep both or drop both: in your latest `Other_terms.tex`, Row II part 1 is commented out but Row V $2b$ UV is not.

2 Notation

$$\mathrm{LO}[f] \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\, f\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[U^{e'c}(y') - U^{e'c}(x')\Big]\,[U^{ec}(y) - U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x), \tag{2.1}$$

$\mathrm{LO} \equiv \mathrm{LO}[1]$. $\mathrm{LO}_\xi[f]$ is the same with $e^{-ik\cdot(y'-y)} \to e^{-ik\cdot(y'-y)\frac{1}{\xi}}$, so $\mathrm{LO}_\xi = \frac{1}{\xi}\frac{d^3N_{\mathrm{LO}}}{d^3k}(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi})$ ($d^3k = dk^+d^2\mathbf{k}$).

$$P^{(r)}_{gg,+}(\xi) = 2N_c\Big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\Big], \qquad P_{gg}(\xi) = P^{(r)}_{gg,+}(\xi) + \frac{\beta_0}{2}\,\delta(1-\xi). \tag{2.2}$$

Here $\frac{1}{(1-\xi)_+}$ is the plus distribution of $[0,1]$. On $\xi \geq a$:

$$\int_a^1 d\xi\,\frac{g(\xi)}{(1-\xi)_+} = \int_a^1 d\xi\,\frac{g(\xi)-g(1)}{1-\xi} + g(1)\log(1-a).$$

$$H_4 \equiv \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\Big[U^{e'c}(z) - U^{e'c}(x')\Big]\,[U^{ec}(z) - U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x). \tag{2.3}$$

3 Step by step

Step 0: your row, line by line.

$$(L1)\quad \frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon} - d_\perp\,\gamma_E - \frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right) - 1\right]$$

$$\left[\int_{\frac{k^+}{\vee}}^1 \frac{d\xi}{\xi^2}\, e^{-ik\cdot(y'-y)\,\frac{1}{\xi}} \left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\xi+\frac{\xi^2}{(1-\xi)_+}\right]\right] \times \Big[U^{e'c}(y')-U^{e'c}(x')\Big] \times \left[U^{ec}(y)-U^{ce}(x)\right]\, \rho^{e'}(x')\rho^{e}(x)$$

$$(L2)\quad -\frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'} \frac{(x-y)\cdot(x'-y)}{(x-y)^2\,(x'-y)^2}\left[-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right]$$
$$\left[\int_{\frac{k^+}{\vee}}^1 \frac{d\xi}{\xi^2}\, e^{-ik\cdot(y'-y)\,\frac{1}{\xi}} \left[\frac{(1-\xi)}{\xi}\right]\right] \times \Big[U^{e'c}(y)-U^{e'c}(x')\Big] \times \left[U^{ec}(y)-U^{ce}(x)\right]\, \rho^{e'}(x')\rho^{e}(x)$$

$$(L3)\quad -\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}$$
$$\Big[U^{e'c}(z)-U^{e'c}(x')\Big] \times \left[U^{ec}(z)-U^{ce}(x)\right]\, \rho^{e'}(x')\rho^{e}(x)$$

$$(L4)\quad +\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.$$
$$\left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\Big[U^{e'c}(y')-U^{e'c}(x')\Big] \times \left[U^{ec}(y)-U^{ce}(x)\right]\, \rho^{e'}(x')\rho^{e}(x)$$

$$(L5)\quad +\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]\times$$
$$\Big[U^{e'c}(y')-U^{e'c}(x')\Big] \times \left[U^{ec}(y)-U^{ce}(x)\right]\, \rho^{e'}(x')\rho^{e}(x) \tag{3.1}$$

Step 1: units of LO. $\frac{g^2}{4\pi^2}=\frac{\alpha_s}{\pi}$. The x,y,x',y' integrands of (L1) and (L4) are those of (2.1), (L1) with $e^{-ik\cdot\Delta}\to e^{-ik\cdot\Delta/\xi}$. With $\mathcal{B}_\Delta$ your bracket:

$$(\text{L1})=\frac{\alpha_s N_c}{\pi}\int_a^1\frac{d\xi}{\xi^2}\Big[\xi(1-\xi)+\frac{1-\xi}{\xi}+\xi+\frac{\xi^2}{(1-\xi)_+}\Big]_{\text{yours}}\text{LO}_\xi[\mathcal{B}_\Delta],\qquad (\text{L4})=\frac{\alpha_s N_c}{\pi}\log(1-a)\,\text{LO}[\mathcal{B}_\Delta]. \tag{3.2}$$

Step 2: the bracket. With $d_\perp=2-\epsilon$,

$$\mathcal{B}_\Delta=-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{\Delta^2\mu^2}{4}-1=-\frac{2}{\epsilon}+1-2\gamma_E-\log\frac{\Delta^2\mu^2}{4}-1+O(\epsilon)=-\frac{2}{\epsilon}-\ell_\Delta(\mu)+O(\epsilon). \tag{3.3}$$

The -1 cancels the $+1$ of $-\frac{d_\perp}{\epsilon}$. The $O(\epsilon)$ multiplies a finite integral and drops out.

Step 3: (L4) completes the plus distribution. Your prescription, $\int\frac{d\xi}{\xi^2}\frac{\xi^2}{(1-\xi)_+}G=\int\frac{d\xi}{1-\xi}[G(\xi)-G(1)]$, has no $\log(1-a)$. On $[a,1]$:

$$\int_a^1\frac{d\xi}{\xi^2}\Big[\xi+\frac{\xi^2}{(1-\xi)_+}\Big]_{\text{yours}}G=\int_a^1\frac{d\xi}{\xi^2}\frac{\xi}{(1-\xi)_+}G-\log(1-a)\,G(1),$$

with $\text{LO}_{\xi=1}=\text{LO}$ (`other_check/checks.py`, item 3). (L4) adds the missing term:

$$(\text{L1})+(\text{L4})=\frac{\alpha_s}{2\pi}\int_a^1\frac{d\xi}{\xi^2}\,P^{(r)}_{gg,+}(\xi)\,\text{LO}_\xi\Big[-\frac{2}{\epsilon}-\ell_\Delta(\mu)\Big]. \tag{3.4}$$

Step 4: split $P^{(r)}_{gg,+}$. By (2.2), $P^{(r)}_{gg,+}=P_{gg}-\frac{\beta_0}{2}\delta(1-\xi)$:

$$(\text{L1})+(\text{L4})=\underbrace{\frac{\alpha_s}{2\pi}\int_a^1\frac{d\xi}{\xi^2}\,P_{gg}(\xi)\,\text{LO}_\xi\Big[-\frac{2}{\epsilon}-\ell_\Delta(\mu)\Big]}_{\text{DGLAP}}+\underbrace{\frac{\alpha_s\beta_0}{4\pi}\,\text{LO}\Big[\frac{2}{\epsilon}+\ell_\Delta(\mu)\Big]}_{\text{running coupling}}. \tag{3.5}$$

Step 5: DGLAP. In the first term, $\int_a^1\frac{d\xi}{\xi^2}P\,\text{LO}_\xi=\int_{k^+/\vee}^1\frac{d\xi}{\xi^3}P(\xi)\frac{d^3N_{\text{LO}}}{d^3k}(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi})$: a gluon of momentum $\frac{k}{\xi}$, produced at LO, turns into one with the fraction ξ. For a hadron,

$$\frac{d^3N_h}{d^3k}=\int_{k^+/\vee}^1\frac{d\zeta}{\zeta^3}\,D^{\text{bare}}_{h/g}(\zeta)\,\frac{d^3N_g}{d^3k}\Big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\Big). \tag{3.6}$$

- With $z=\zeta\xi$, the region $a\le\zeta\le1$, $\frac{a}{\zeta}\le\xi\le1$ is $a\le z\le\zeta\le1$. This needs the plus distribution of Step 3, i.e. (L4).
- The pole becomes $\frac{\alpha_s}{2\pi}(-\frac{2}{\epsilon})\int_a^1\frac{dz}{z^3}\frac{d^3N_{\text{LO}}}{d^3k}(\frac{k}{z})\,[P_{gg}\otimes D](z)$, with $[P\otimes D](z)=\int_z^1\frac{d\zeta}{\zeta}P(\frac{z}{\zeta})D(\zeta)$ (`other_check/checks.py`, item 5: 2×10^{-12}).
- In $\overline{MS}$ at μ_F it is removed by

$$D^{\text{bare}}_{h/g}(z)=D_{h/g}(z,\mu_F)+\frac{\alpha_s}{2\pi}\Big[\frac{2}{\epsilon}+\log\frac{\mu^2}{\mu_F^2}\Big][P_{gg}\otimes D_{h/g}](z,\mu_F),\qquad \mu_F^2\frac{\partial D_{h/g}}{\partial\mu_F^2}=\frac{\alpha_s}{2\pi}P_{gg}\otimes D_{h/g}. \tag{3.7}$$

What the DGLAP term leaves in the gluon cross section ($\ell_\Delta(\mu) = \ell_\Delta(\mu_F) + \log\frac{\mu^2}{\mu_F^2}$):

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\rm DGLAP} = -\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1 \frac{d\xi}{\xi^2}\,P_{gg}(\xi)\,{\rm LO}_\xi\big[\ell_\Delta(\mu_F)\big].} \tag{3.8}$$

Its μ_F dependence cancels that of $D_{h/g}$ through (3.7).

Step 6: running coupling. LO is $\propto g^2$. In $\overline{MS}$, $g^2 \to g^2(\mu_R)\big[1 - \frac{\alpha_s\beta_0}{4\pi}(\frac{2}{\epsilon} + \log\frac{\mu^2}{\mu_R^2})\big]$ adds $-\frac{\alpha_s\beta_0}{4\pi}(\frac{2}{\epsilon} + \log\frac{\mu^2}{\mu_R^2})$LO. Added to the second term of (3.5):

$$\frac{\alpha_s\beta_0}{4\pi}{\rm LO}\Big[\frac{2}{\epsilon} + \ell_\Delta(\mu)\Big] - \frac{\alpha_s\beta_0}{4\pi}\Big(\frac{2}{\epsilon} + \log\frac{\mu^2}{\mu_R^2}\Big){\rm LO} = \boxed{\frac{\alpha_s\beta_0}{4\pi}\,{\rm LO}\Big[\log\frac{(y'-y)^2\mu_R^2}{c_0^2}\Big].} \tag{3.9}$$

The pole and μ cancel. With $\alpha_s(Q) = \alpha_s(\mu_R)\big[1 - \frac{\alpha_s\beta_0}{4\pi}\log\frac{Q^2}{\mu_R^2}\big] + O(\alpha_s^3)$ and $Q = \frac{c_0}{|y'-y|}$:

$${\rm LO}\big|_{\alpha_s(\mu_R)} + \frac{\alpha_s\beta_0}{4\pi}{\rm LO}\big[\ell_\Delta(\mu_R)\big] = {\rm LO}\Big[\alpha_s\Big(\frac{c_0}{|y'-y|}\Big)\Big] + O(\alpha_s^3). \tag{3.10}$$

- This is the running of the coupling of LO. The scale is the separation $|y'-y|$ of the measured gluon in the amplitude and in the conjugate amplitude, which is conjugate to $\mathbf{k}$; μ_R has dropped out.
- If you prefer a fixed scale: (3.9) $= \frac{\alpha_s\beta_0}{4\pi}\log\frac{\mu_R^2}{\mu_F^2}{\rm LO} + \frac{\alpha_s\beta_0}{4\pi}{\rm LO}[\ell_\Delta(\mu_F)]$. The first term is the renormalization-group term of `Other_terms_analysis.pdf`; the second combines with (3.8) into $-\frac{\alpha_s}{2\pi}P^{(r)}_{gg,+} \otimes {\rm LO}[\ell_\Delta(\mu_F)]$. The two groupings are identical (sympy check of the pole and μ bookkeeping in `make_note.py`).

Step 7: (L2)+(L3). In (L2), y' appears only in $e^{-ik\cdot\Delta/\xi}$ and the log, and $\frac{d_\perp}{2} \to 1$ (no pole).

- $\int_{y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\big[-\log\frac{(y'-y)^2\mu^2}{4}\big] = \frac{4\pi\xi^2}{\mathbf{k}^2}$ for $\mathbf{k}\neq 0$; the μ term multiplies $\delta^{(2)}(\mathbf{k})$ and drops out.
- $\int_a^1 \frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}\xi^2 = \log\frac{\vee}{k^+} - (1 - \frac{k^+}{\vee})$.
- $\int_{w,w'} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2} = \frac{(2\pi)^2}{\mathbf{k}^2}$.

With $y \to z$ the prefactors agree, so

$$({\rm L2}) = -\Big[\log\frac{\vee}{k^+} - \Big(1 - \frac{k^+}{\vee}\Big)\Big]H_4, \qquad ({\rm L3}) = -\Big(1 - \frac{k^+}{\vee}\Big)H_4, \qquad ({\rm L2}) + ({\rm L3}) = -\log\frac{\vee}{k^+}\,H_4. \tag{3.11}$$

This cancels the $+\log\frac{\vee}{k^+}H_4$ of the BFKL list of this row and makes (3.8) finite for $\vee \to \infty$.

Step 8: (L5). The linear-polarization line has no pole, so $d_\perp = 2$ in it.

- The Δ angular average of its bracket is zero, so it is not DGLAP; its Fourier transform carries $\frac{2k^ik^j}{\mathbf{k}^2} - \delta^{ij}$.
- Its measure is $d\xi$, not $\frac{d\xi}{\xi^2}$, so it is finite for $\vee \to \infty$.

4 Result: the row after DGLAP and running coupling

After $D_{h/g}(\zeta, \mu_F)$ takes the P_{gg} pole and the coupling counterterm takes the β_0 pole, your row is ($\epsilon \to 0$ exact in what is finite)

$$\begin{aligned}\frac{d^3N^{\rm G2\,IV}}{d^3k} = &-\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,P_{gg}(\xi)\,{\rm LO}_\xi\big[\ell_\Delta(\mu_F)\big] \qquad \text{(DGLAP, finite part)}\\ &+\frac{\alpha_s\beta_0}{4\pi}\,{\rm LO}\big[\ell_\Delta(\mu_R)\big] \qquad \text{(running coupling: } \alpha_s(\mu_R) \to \alpha_s(c_0/|y'-y|) \text{ in LO)}\\ &-\log\frac{\vee}{k^+}\,H_4 \;+\; ({\rm L5})_{d_\perp=2}.\end{aligned} \tag{4.1}$$

In your notation:

$$\begin{aligned}\frac{d^3N^{\rm G2\,IV}}{d^3k} = &\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[-2\gamma_E - \log\frac{(y'-y)^2\mu_F^2}{4}\Big]\\ &\times\Big[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\Big[\xi(1-\xi) + \frac{(1-\xi)}{\xi} + \xi + \frac{\xi^2}{(1-\xi)_+}\Big] + \Big(\log\Big(1-\frac{k^+}{\vee}\Big) + \frac{11}{12}\Big)e^{-ik\cdot(y'-y)}\Big]\\ &\times\Big[U^{e'c}(y') - U^{e'c}(x')\Big]\,[U^{ec}(y) - U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x)\end{aligned}$$

$$
\begin{aligned}
&+\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\frac{11}{12}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[2\gamma_E+\log\frac{(y'-y)^2\mu_R^2}{4}\right]\\
&\quad\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\left[U^{ec}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)\\
&-\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\\
&\quad\times\left[U^{e'c}(z)-U^{e'c}(x')\right]\left[U^{ec}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)\\
&+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]\\
&\quad\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\left[U^{ec}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x). \qquad (4.2)
\end{aligned}
$$

- Lines 1–3: the DGLAP term (3.8). With the prefactor $\frac{g^2N_c}{4\pi^2}=\frac{\alpha_s}{2\pi}2N_c$, the $\frac{11}{12}$ is $\frac{\beta_0}{2}\cdot\frac{1}{2N_c}$, the $\delta(1-\xi)$ of P_{gg}.
- Lines 4–5: the running coupling (3.9), $\frac{\alpha_s\beta_0}{4\pi}=\frac{g^2N_c}{4\pi^2}\cdot\frac{11}{12}$.
- Lines 6–7: (L2)+(L3).
- Lines 8–9: (L5) with $d_\perp=2$.

The two $\frac{11}{12}$ terms cancel when $\mu_R=\mu_F$; their sum is always $\frac{\alpha_s\beta_0}{4\pi}\log\frac{\mu_R^2}{\mu_F^2}$LO.